BSCS/BSSE/BSDS/BSAI FINAL YEAR PROJECT PROPOSAL

ZehenSync AI: A Hybrid Multimodal System for Behavioral Screening

Term of Registration: Fall 2026

Particulars of the students:

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Project Type:

☒ **Research based**	☐ **Hardware based/Embedded**	☐ **Game based**
☐ **Software Development**	☒ **Artificial Intelligence (AI)**	☐ **Mobile Application**
☒ **Web Application**	☐ **Robotics**	☐ **Database**
☐ **Cyber Security**	☐ **Other: _____**	

Faculty of Information Technology & Computer Science

University of Central Punjab

Project Title

ZehenSync AI: A Hybrid Multimodal System for Behavioral Screening

Project Advisor

Iqra Tariq

Acceptance of Project Idea

Students will be required to defend their idea before the scrutiny committee (SC) which has the authority to accept or reject the project idea. The decision taken by the SC will be final and cannot be challenged.

Similarly, in phase wise evaluations, the marks awarded by the evaluators will be considered final. No excuses on their skill, relevancy, competency or biasedness will be acceptable as an absolve.

Supervisory Meetings

A group is required to hold at least two meetings per month and maintain meeting minutes. Missing two consecutive meetings without any notice may lead to withdrawal of the project.

Advisor's Consent

I Prof./Dr./Mr./Ms. _____ am willing to guide these students in all phases of above-mentioned project as advisor. I have carefully seen the Title and description of the project and believe that it is of an appropriate difficulty level for the number of students named above.

Note:
Advisor can't be changed and the duration for completion of the Project is 2 regular semesters (approx.) from the date of Registration of the Project.

Signatures and Date

Advisor

Sr. #	Name	Registration #	Signatures
1	SYED RAZA ABBAS ZAIDI	L1F23BSAI0011	
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1. Abstract

When we go for a routine medical checkup, our mental health is almost always ignored. This leaves a massive gap in healthcare, as we lack objective measurement tools and AI systems with deep psychological training. Usually, doctors rely entirely on what patients tell them, which leads to the "blank slate" problem: deeply distressed patients often struggle to start from scratch and explain complex feelings in an open-ended conversation. To solve this, our project, ZehenSync AI, introduces "Behavioral Medical Imaging." The system uses standard video like a medical scan to track physical signs, such as how often a person blinks and where they are looking. At the same time, it analyzes their voice for speech patterns, pauses, and stutters. These physical signs are sent to a highly specialized AI that has been specifically trained on official psychology manuals, like the DSM-5, and real clinical cases. ZehenSync AI features three distinct modules. The Patient Module uses "Guided Volition" to gently lead patients through structured, easy to answer screenings, eliminating the pressure of an empty text box. The Practitioner Module is designed for psychologists, allowing them to review the objective AI generated reports. Doctors can use this tool to prioritize urgent cases, forward patients to the right specialists, and even consult the AI for diagnostic support. Additionally, the Scholar Module provides a dedicated space for students and academic research. Safety is the absolute priority. If the system detects any signs of a severe crisis or emergency, it instantly overrides the normal screening process and routes the patient directly to the Umang Pakistan helpline for immediate, real-world help. Ultimately, ZehenSync AI does not replace human psychologists; it acts as a smart assistant providing evidence-based data so doctors can catch mental health issues earlier and more accurately.

2. Introduction of the Project

ZehenSync AI is an innovative mental health assessment platform operating within Digital Phenotyping and Behavioral Informatics. According to data released by the World Health Organization (WHO) in late 2025, over one billion individuals worldwide are currently living with a mental health condition, representing a staggering 23.4% of the global adult population.[1] Historically, mental health has been a subjective, qualitative field, and during routine physical checkups, our psychological well being is often completely overlooked. Generic AI chatbots attempt to fill this gap but fail due to the "Blank Slate Problem" waiting passively for vulneerable users who may not know how to articulate their struggles.

ZehenSync AI solves this by shifting to "Guided Volition," actively leading users through proven psychological evaluations. It captures visual and vocal cues, treating them as measurable "medical Regions of Interest" with the objectivity of a radiologist reading an X-ray. The motivation for this project stems from recognizing this critical healthcare gap and the limitations of current conversational AI. Leveraging a technical background in machine learning and data visualization, the goal is to create a traceable, logical system. By translating raw behavioral signals into reliable data, ZehenSync AI creates a structured, ground truth record that human therapists and doctors can easily review and trust.

ZehenSync AI implements a specialized RBAC (Role-Based Access Control) system featuring three distinct entry points:

- Patient Role: Utilizes "Guided Volition" to navigate structured screenings, removing the "blank slate" anxiety.
- Practitioner Role: Designed for psychologists to review AI-generated reports, prioritize urgent cases, and access diagnostic support.
- Student Role: Acts as a "Scholar Module," providing a zero-risk environment for clinical simulations and DSM-5 research.

As of 2026, the behavioral health workforce faces an 83% burnout rate, with demand for services consistently outstripping provider supply.[3] ZehenSync AI acts as a diagnostic co-pilot, helping in administrative burdens and providing objective evidence for complex cases. The platform uses a system that gives different access to three types of people. The Patient is the main user who follows an easy, step-by-step process called "Guided Volition" to answer questions without feeling overwhelmed. The Practitioner role is for doctors and psychologists, giving them a safe way to check AI reports and quickly find the most urgent cases. Lastly, the Student role uses the "Scholar Module" to study clinical cases and practice research using official psychology manuals.

3. Problem Statement & Gap Analysis

Current medical protocols rely heavily on "patient-doctor spontaneity," wherein the burden of identifying psychological distress falls upon the patient's ability to self-report. The result is a failure of screening that borders on negligence; recent data from 2024 and 2025 indicates that standard primary care visits miss over 40% of cases involving clinical depression and anxiety.[2] Current mental health tools are fragmented; eye-tracking and voice analysis exist as isolated technologies providing raw data without clinical context. Furthermore, general-purpose AI models (GPT, Llama) lack specialized training in peer-reviewed psychological literature like the DSM-5-TR, limiting their ability to interpret complex patient behavior accurately. There is a pressing need for a cohesive system that detects behavioral indicators through deep learning and explains them through a medically expert language model. Unlike competitors like Wysa or Ellipsis, which focus on single modalities or heuristic-based chat, ZehenSync AI provides a multimodal SDK and a fine-tuned reasoning engine with traceable citations. This systemic oversight creates a massive gap in healthcare delivery, exacerbated by a critical shortage of objective measurement tools and a lack of artificial intelligence systems with deep, specialized psychological training.

Mental Health Indicator (2024-2025)	Statistic	Source
Global Prevalence of Mental Disorders	1.0 Billion +	WHO 2025
U.S. Adults with Mental Health Conditions	61.5 Million	NSDUH 2024
Percentage Receiving No Treatment	~48%	NSDUH 2024
Annual Cost of Untreated Mental Illness	$477.5 Billion	Trilliant Health 2024
Primary Care Miss Rate for Depression	42.3%	AMA 2023
Primary Care Miss Rate for Anxiety	42.7%	AMA 2023

4. Proposed Solution / Product Concept

ZehenSync AI is a hybrid multimodal system combining computer vision, speech analysis, and a fine-tuned psychology expert LLM. A key feature of the architecture is the Role-Based Access Control (RBAC) system, which creates three distinct user pathways: the Patient Role for guided self-assessments, the Practitioner Role for clinical review and reporting, and the Student Role for academic research and training.

- Visual Engine: Uses YOLO v12 and Vision Transformers for precise tracking of ocular biomarkers.
- Speech Engine: Employs Whisper to analyze acoustic markers like pauses and disfluencies.
- Cognitive Explanation Engine: A fine-tuned Gemma 3 or Qwen 3 model using RAG to synthesize data with clinical literature.
- Guided Volition: A UI strategy using clickable, validated options to eliminate data noise.

This solution provides psychologists with objective metrics, reducing administrative burden and cognitive load during sessions

5. Objectives & Target Market

Objectives: To create a Psychology-Expert LLM (Psyche-Corpus), implement a visual behavioral engine, and build a multimodal fusion pipeline with RAG for evidence-based reporting.

- Create a Psychology-Expert LLM: Develop the "Psyche-Corpus" by fine-tuning models like Gemma 3 or Qwen 3 on official medical literature such as the DSM-5-TR.

- Establish a Visual Behavioral Engine: Implement YOLO v12 to track ocular biomarkers, including blink rates and gaze stability.
- Implement RBAC Security: Ensure high data privacy by separating user permissions into specific forks for Patients, Practitioners, and Students.
- Automate Clinical Reporting: Build a system that generates PDF screening reports to reduce the administrative workload for doctors.
- Integrate Emergency Protocols: Maintain a 100% success rate for a "Hard Stop" protocol that instantly routes high-risk users to crisis helplines like Umang Pakistan.
- Validate Scoring Logic: Achieve 100% test coverage for all clinical scoring algorithms, such as the PHQ-9 and GAD-7, to ensure accurate results.

Target Market: University psychology departments, specialized diagnostic centers, and mental healthcare researchers. The Practitioner Module is designed to empower, not replace human psychologists.

- University Psychology Departments: For use as a teaching tool and for conducting behavioral informatics research.
- Specialized Diagnostic Centers: To provide doctors with objective data that supports traditional clinical interviews.
- Mental Healthcare Researchers: Who require a multimodal SDK to analyze large sets of behavioral data.
- General Healthcare Clinics: To help identify mental health issues during routine physical checkups where psychological health is often ignored.
- Academic Students: Providing a zero-risk environment for students to simulate clinical evaluations and study symptom clusters

Value Add: Increases accessibility to early screening and provides students with a zero-risk environment for clinical simulations

6. Customer & User Research

The primary beneficiaries include Clinicians, who require automated observation tools to focus on therapeutic relationships, and Students, who need educational tutors for DSM-5 research.

- Research Findings: Insights from the Clinical Psychology Department helped establish "Ground Truth" protocols essential for clinical validity.
- Collaboration: External partners, including local mental health centers and the UCP Clinical Psychology Department, provide access to clinical data for validation.
- Key Discovery: The discovery of the "Blank Slate Problem" led to our strategic pivot toward the "Guided Volition" model.

7. Tools and Technologies

Frontend: Next.js/Flask/Django (Patient/Scholar/Practitioner Fork).

Backend: FastAPI, WebSockets for real-time interaction.

Database: PostgreSQL (User data), ChromaDB (Vector store for RAG).

AI/ML: PyTorch, TensorFlow, YOLO v12, MediaPipe, Whisper, Gemma 3/Qwen 3.

Cloud: Google Cloud Platform (GCP) , Alibaba Cloud or AWS.

8. **Expected Outcomes /KPI's**

Product Outcomes: A functional web platform (ZehenSync Dash) and a fine-tuned Psyche-Expert LLM.

- Functional Web Platform: The delivery of a secure, cloud-deployed portal called ZehenSync Dash that handles real-time video and audio analysis.
- Psyche-Expert LLM: A fine-tuned AI model (Gemma 3 or Qwen 3) that provides clinical reasoning and evidence-based citations using a RAG pipeline.
- Multimodal SDK: A technical framework that works with standard webcams to track ocular biomarkers and analyze speech patterns.

Customer Outcomes: Significant reduction in clinician administrative burden and enhanced screening accessibility for the public.

- Reduced Administrative Burden: A significant decrease in the manual paperwork and observation time required by clinicians.
- Enhanced Accessibility: Improved early screening for the public by integrating mental health checks into routine medical checkups.
- Objective Reporting: The generation of automated PDF screening support reports that map non-verbal behaviors to established clinical criteria.

KPIs: Achievement of clinical reasoning benchmarks higher than general models and 100% success rate for the "Hard Stop" crisis protocol.

- Clinical Reasoning Benchmarks: The specialized AI model must achieve higher accuracy in psychology-related tasks compared to general-purpose AI models.
- Crisis Protocol Reliability: A 100% success rate for the "Hard Stop" emergency protocol to ensure high-risk users are instantly connected to crisis help.
- System Performance: The platform must maintain a system latency of less than 2 seconds when processing and scoring questionnaires under peak load.
- Logic Verification: Achieving a 100% pass rate for all clinical scoring algorithms (like PHQ-9 and GAD-7) during unit testing.
- Traceability: Ensuring every clinical requirement is mapped to a specific test case with zero "orphan" requirements.

9. **Completeness Criteria**

Project success is defined by four milestones:

1. Core Architecture: Foundation of RBAC and the initial YOLO 12 vision baseline.
2. Engine Integration: Completion of the Speech and Vision pipelines with real-time WebSocket communication.
3. The Brain: Fine-tuning the LLM on the Psyche-Corpus and implementing the RAG citation sidebar.

4. Clinical Validation: Testing the "Guided Volition" flow and generating PDF screening reports.

10. Challenges

Technical: Managing time for complex FYP tasks and improving skills in emerging tools like YOLO 12 and Vision Transformers.

Environmental: Behavioral metrics accuracy is sensitive to user camera quality and lighting.

Consistency: Maintaining focus through the two-semester timeline by adhering to the Agile sprint framework.

11. Project Success Criteria

The final result of ZehenSync AI is a comprehensive AI-driven software platform that serves as both a clinical screening tool and an academic research assistant. It consists of three tangible outcomes:

- A Web-Based Application: A secure portal where patients undergo guided assessments and clinicians review objective behavioral reports.
- A Fine-Tuned Psychology LLM: A specialized model optimized for clinical reasoning and evidence-based citations.
- A Multimodal SDK: A technical framework for real-time video and audio analysis using standard webcams.

The project is successful if it provides a traceable logic chain that accurately maps non-verbal behaviors to established clinical criteria.

12. Related Work / Literature Survey / Literature Review

ZehenSync AI builds upon existing specialized technologies:

- Voice Biomarkers: Platforms like Ellipsis Health and Winterlight Labs detect depression through acoustic features.
- Ocular Tracking: Hardware-based tools like Tobii Pro offer high precision but require expensive specialized equipment.
- Chatbots: Wysa and Woebot provide CBT support but lack deep multimodal integration and specialized clinical citation capabilities.
- Our Difference: Unlike these fragmented solutions, ZehenSync AI provides a Multimodal SDK that combines text, audio, and video using standard hardware (webcams), supported by a fine-tuned psychology-expert LLM with traceable Source_ID logic.

13. Project Plan / Project Schedule / Project Timetable / Project Calendar

The project plan for ZehenSync AI follows an Agile framework organized into four-week sprints to ensure iterative testing of multimodal features and clinical validity. The schedule spans from the starting date of April 30, 2026, to the projected completion in March 2027.

Project Schedule:

Week	Phase	Key Activities & Subtasks	Resources Used
1-4	Sprint 1: Foundations	Setup Next.js/FastAPI; implement RBAC; develop 'Guided Volition' intake logic.	Next.js, PostgreSQL
5-8	Sprint 2: Vision Core	Baseline YOLO 12 tracking; extract blink rate and gaze stability metrics.	YOLO 12, MediaPipe
9-12	Sprint 3: Speech Core	Integrate Whisper for STT; extract acoustic markers (pauses, disfluencies).	OpenAI Whisper, Librosa
13-16	Sprint 4: Psyche-Corpus	Curate specialized psychology dataset; index DSM-5 and case studies in ChromaDB.	DSM-5, ChromaDB
17-20	Sprint 5: LLM Fine-Tuning	Fine-tune Gemma 3/Qwen 3 using LoRA/QLoRA on specialized datasets.	GCP / Alibaba Cloud
21-24	Sprint 6: Fusion & RAG	Implement RAG pipeline to combine visual/vocal metrics with LLM reasoning.	LangChain, FastAPI
25-28	Sprint 7: Safety Logic	Build 'Hard Stop' crisis protocol; trigger emergency resources (Umang Pakistan).	Umang API, Python
29-32	Sprint 8: Clinical Reporting	Develop automated PDF screening support report generator for clinicians.	ReportLab, Next.js
33-36	Sprint 9: Validation	Conduct 'Ground Truth' testing and logic chain verification with advisors.	Clinical Advisors
37-40	Sprint 10: Finalization	Deploy to Cloud; final bug fixes; complete FYP documentation and defense.	GCP, MS Word

Use of Resources per Subtask:

- Technical Infrastructure: Google Cloud Platform (GCP) or Alibaba Cloud provide high-performance computing (GPUs/TPUs) for LLM fine-tuning and YOLO 12 inference.
- Data Sources: The Psyche-Corpus, containing DSM-5 criteria and academic papers, acts as the primary knowledge base for the RAG system.
- Development Tools: Next.js is used for the frontend user fork (Patient vs. Scholar vs. Practitioner), while FastAPI orchestrates the backend multimodal engines.

- External Support: Local mental health organizations and the UCP Clinical Psychology Department provide the clinical data needed for validation studies.

Monitoring and Evaluation:

- Progress Tracking: We utilize a Requirements Traceability Matrix (RTM) to ensure every feature, such as objective ocular metrics (NL-03), maps back to specific design components and test cases.
- Advisor Meetings: As per university protocol, the group holds at least two meetings per month with advisor Iqra Tariq to review meeting minutes and technical progress.
- Academic Oversight: The Scrutiny Committee (SC) evaluates the project in phases, ensuring it meets academic standards and ethical guidelines.

14. References/Bibliography

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7. Mate, J., & Micheal, D. (2026). Integrating Computer Vision and Behavioral Data for AI-Based Autism Screening. *ResearchGate*.